Physical AI Closed Loop System for Future OEE Forecasting using Artificial Neural Networks

Industrial IoT & AI/ML Project

June 17, 2026

1 Manufacturing Efficiency and OEE Fundamentals

1.1 The Need for Measuring Manufacturing Efficiency

Manufacturing productivity requires a standardised metric to quantify losses and benchmark performance. Overall Equipment Effectiveness (OEE) provides a dimensionless number (0–100%) capturing three independent dimensions: Availability, Performance, and Quality.

1.2 The Three OEE Components

OEE decomposes into Availability, Performance, and Quality (ISO 22400-2). Each addresses a distinct loss category:

Planned Production Time ↓
Availability Loss (Breakdowns, Setup) → **Running Time** ↓
Performance Loss (Slow cycles, Stops) ↓
Quality Loss (Defects, Rework) ↓
Fully Productive Time

Availability measures the ratio of actual operating time to planned production time:

$$\text{Availability} = \frac{\text{Operating Time}}{\text{Planned Production Time}} \times 100 \tag{1}$$

Availability losses include equipment breakdowns, unscheduled maintenance, and changeover time. In our lathe dataset, availability typically ranges from 70% to 100%, with downtime events injecting sudden drops followed by gradual recovery.

Performance measures how fast the machine runs compared to its designed ideal speed:

$$\text{Performance} = \frac{\text{Ideal Cycle Time} \times \text{Total Parts}}{\text{Operating Time}} \times 100 \tag{2}$$

Performance losses include slow running (below design speed), minor stops, and operator inefficiency. In our system, performance is directly coupled to the motor RPM (300–500 range), with higher speeds yielding better performance but potentially degrading quality.

Quality measures the proportion of good parts out of total produced:

$$\text{Quality} = \frac{\text{Good Parts}}{\text{Total Parts}} \times 100 \tag{3}$$

Quality losses include scrap, rework, and yield loss. In our synthetic dataset, quality degrades gradually over each 8-hour shift and resets at shift boundaries, simulating tool wear patterns.

1.3 OEE Calculation and Benchmarks

These three factors combine multiplicatively, meaning a deficiency in any single component directly compresses the final OEE:

$$\text{OEE} = \frac{\text{Availability} \times \text{Performance} \times \text{Quality}}{10000} \tag{4}$$

The division by 10,000 arises because all three inputs are percentages (0–100). World-class OEE benchmarks established by the Japan Institute of Plant Maintenance (JIPM) are:

Metric	**World-Class**	**Typical Average**
Availability	90%+	75–85%
Performance	95%+	60–75%
Quality	99%+	90–97%
OEE	**85%+**	**55–65%**

Table 1: OEE benchmarks by class (source: JIPM / ISO 22400-2).

An OEE of 85% is considered world-class; most manufacturing plants operate between 55% and 65%. Each 1% improvement in OEE directly translates to increased throughput without capital expenditure.

1.4 Why Predict OEE?

Traditional OEE measurement is *reactive* – it tells operators what already happened. This project develops a *predictive* OEE system that forecasts future OEE at multiple horizons (30 minutes to 8 hours), enabling proactive decision-making. The key insight is that OEE is not a random variable; it evolves deterministically based on machine state (RPM, downtime, quality trends), making it suitable for data-driven forecasting with artificial neural networks.

2 Digital Twin Architecture and Data Acquisition

2.1 Digital Twin Concept

A Digital Twin is a virtual representation of a physical manufacturing system that mirrors its real-time state through continuous data synchronisation. Unlike static simulations, a Digital Twin maintains bidirectional communication with the physical asset: sensor data flows from the machine to the digital model, and control signals (such as RPM setpoints) flow back from the digital model to the machine.

2.2 System Architecture

The closed-loop pipeline: **Factory I/O** (conveyor 3–18 mm/s) → **TIA Portal PLC** (S7-1200, ISO-on-TCP, IP 10.11.15.193) → **Node-RED** (poll S7, compute A/P/Q/OEE, write MySQL) → **MySQL** (`lathe.manufacture_ai_data`) → **FastAPI + ANN** (predict/optimise) → **Node-RED Dashboard** (gauges) → **VFD** (Hz = RPM/10) → back to Factory I/O.

2.3 Control Variable: Feed Rate and RPM

The manufacturing process centres on a single conveyor feed mechanism. The physical control variable is the `feed_rate_mmps`, operating in the range 3 mm/s to 18 mm/s. This feed rate is converted to motor RPM through a linear mapping:

$$\text{RPM} = 300 + (\text{feed\_rate\_mmps} - 3) \times 13.33 \tag{5}$$

Thus, 3 mm/s corresponds to 300 RPM and 18 mm/s to 500 RPM. For VFD (Variable Frequency Drive) integration, the frequency command is:

$$\text{Hz} = \frac{\text{RPM}}{10} \tag{6}$$

This gives a 30–50 Hz range compatible with standard industrial VFDs. The feed rate influences all three OEE components:

- **Availability**: higher feed rates increase mechanical stress and downtime probability.
- **Performance**: directly proportional to feed rate (more parts per unit time).
- **Quality**: excessive speed increases defect rate (tool chatter, surface finish degradation).

2.4 Data Acquisition via Node-RED

Node-RED connects to the Siemens S7-1200 PLC via `node-red-contrib-s7` (ISO-on-TCP, port 102). At 1-minute intervals it: reads RPM, part count, downtime, rejects; computes A/P/Q/OEE; writes to `lathe.manufacture_ai_data`; calls the FastAPI prediction endpoint; and updates dashboard gauges.

3 Synthetic Time-Series Dataset Generation

3.1 Why Synthetic Data?

The real dataset has only 5 rows – insufficient for deep learning. An initial regression model (Polynomial Degree 2, $R^2 \approx 0.995$) learned only instantaneous state-to-OEE mappings without temporal understanding. Time-series forecasting requires data that: (1) spans multi-hour temporal patterns, (2) has sequential dependencies, (3) includes realistic dynamics (shifts, tool wear, downtime), and (4) has known ground-truth targets for supervised learning.

3.2 Dataset Size Rationale (Why 20,000 Rows?)

The dataset size selection follows from three constraints:

1. Temporal coverage requirement. The longest forecast horizon is 8 hours (480 minutes). To have sufficient training examples for 8-hour predictions, the dataset must span multiple days. At a 1-minute sampling rate, 20,000 rows cover approximately 13.9 days of continuous production – enough for the model to learn daily and multi-day patterns.

2. Parameter-to-sample ratio. Each ANN (Section 4) has 3,073 trainable parameters. A rule of thumb in deep learning is 5–10 training samples per parameter to avoid overfitting. With 15,616 training samples (80% of 19,520), we achieve approximately 5.1 samples per parameter – at the lower bound but acceptable with early stopping and regularisation.

3. Industry precedent. Published OEE forecasting papers (CIRP 2023, MDPI Sensors 2022) use datasets of 3,000–5,000 rows. Our 20,000-row dataset is 4–6 times larger, providing a safety margin for the more complex ANN architecture.

3.3 Data Generation Methodology

The synthetic data generator creates 20,000 rows at 1-minute intervals starting from a known seed distribution:

The simulation models several realistic phenomena:

- **RPM random walk**: mimics operator adjustments and process drift.
- **Downtime events**: 3% probability per minute with exponentially distributed duration (mean 5 min, max 20 min).
- **Quality shift cycle**: resets every 480 minutes (8-hour shift), simulating tool change or shift handover.

Feature	Unit	Range	Generation Method
`rpm`	RPM	300–500	Random walk ± 5, clip at bounds
`availability`	%	70–100	State evolution: drop $\propto$ downtime, recover $+0.1$/step
`performance`	%	30–100	Linear with RPM: RPM/500 $\times$ 100 $\pm$ 3 noise
`quality`	%	60–100	Gradual decay (-0.001/step), reset every 480 steps
`current_oee`	%	15–99	Computed: A $\times$ P $\times$ Q / 10000
`downtime_minutes`	min	0–20	3% event probability, Exp($\lambda = 5$) magnitude

Table 2: Six input features with generation methods and ranges.

- **Availability recovery**: gradual uptick when no downtime occurs, modelling reactive maintenance.

3.4 Target Generation for Multi-Horizon Forecasting

The targets are created by shifting `current_oee` forward in time:

$$\text{target\_oee\_H} = \text{current\_oee}_{t+H} \tag{7}$$

where $H \in \{30, 60, 120, 360, 480\}$ minutes, corresponding to horizons of 30m, 1h, 2h, 6h, and 8h. Shifting creates NaN values for the last H rows, which are dropped. Starting from 20,000 rows, this yields:

$$\text{Usable rows} = 20000 - \max(H) = 20000 - 480 = 19520 \tag{8}$$

The chronological ordering is preserved throughout, and the temporal split (80% train, 10% validation, 10% test) respects time order – no shuffling, preventing data leakage from future to past.

4 Artificial Neural Network Design and Training

4.1 Why an Artificial Neural Network?

Polynomial regression (the baseline model) can capture non-linear relationships between input features and output, but it cannot model the temporal structure implicit in the data. An ANN offers two key advantages:

1. **Hierarchical feature learning**: hidden layers learn progressively more abstract representations of the input space.
2. **Non-linear universal approximation**: a feedforward network with sufficient hidden units can approximate any continuous function (Universal Approximation Theorem, Cybenko 1989).

4.2 Network Architecture

Each horizon model uses an identical architecture:

$$\mathbb{R}^6 \xrightarrow{W_1,b_1} \mathbb{R}^{64} \xrightarrow{\sigma} \mathbb{R}^{64} \xrightarrow{W_2,b_2} \mathbb{R}^{32} \xrightarrow{\sigma} \mathbb{R}^{32} \xrightarrow{W_3,b_3} \mathbb{R}^{16} \xrightarrow{\sigma} \mathbb{R}^{16} \xrightarrow{W_4,b_4} \mathbb{R}^1 \tag{9}$$

where W_i are weight matrices, b_i bias vectors, and σ the ReLU activation function:

$$\text{ReLU}(x) = \max(0, x) \tag{10}$$

ReLU is chosen for hidden layers because it mitigates the vanishing gradient problem, is computationally efficient, and promotes sparse activation. The output layer uses a linear activation (no squashing) since OEE is a continuous regression target in [0, 100].

$$\texttt{Input[6]} \xrightarrow{448+64} \texttt{Dense(64)+ReLU} \xrightarrow{2080+32} \texttt{Dense(32)+ReLU} \xrightarrow{528+16} \texttt{Dense(16)+ReLU} \xrightarrow{17+1} \texttt{Dense(1)+Linear} \rightarrow \textbf{Predicted OEE}$$

Total: **3,073 parameters** per model (6 input features, 3 hidden layers, 1 output).

4.3 Training Pipeline

Loss function: Mean Squared Error (MSE):

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{11}$$

MSE penalises large errors quadratically, which drives the model to avoid large prediction deviations at the cost of accepting many small errors.

Optimiser: Adam (Adaptive Moment Estimation, Kingma and Ba 2014):

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \tag{12}$$

where $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected estimates of the first and second moments of the gradient. Adam combines the benefits of AdaGrad (adaptive learning rates) and RMSProp (gradient normalisation) with momentum. Learning rate $\eta = 0.001$.

Training configuration:

- Chronological split: 80% train (15,616 rows), 10% validation (1,952), 10% test (1,952).
- Feature scaling: StandardScaler (zero mean, unit variance), fitted on training set only.
- Batch size: 32 (mini-batch gradient descent).
- Epochs: 100 maximum, with EarlyStopping (patience=10, restore best weights).
- Regularisation via early stopping prevents overfitting to training noise.

Backpropagation: Gradients propagate backward via the chain rule: $\frac{\partial \mathcal{L}}{\partial W_{ij}^{(l)}} = \frac{\partial \mathcal{L}}{\partial a_j^{(l)}} \cdot \frac{\partial a_j^{(l)}}{\partial z_j^{(l)}} \cdot \frac{\partial z_j^{(l)}}{\partial W_{ij}^{(l)}}$, where $z_j^{(l)} = \sum_i W_{ij}^{(l)} a_i^{(l-1)} + b_j^{(l)}$ is the pre-activation and $a_j^{(l)} = \sigma(z_j^{(l)})$ the post-activation.

5 Future OEE Forecasting Results and Future Scope

5.1 Forecasting Pipeline

The complete prediction pipeline for a single query is:

1. Receive current machine state: RPM, Availability, Performance, Quality, OEE, Downtime.
2. Scale the 6-dimensional input vector using the pre-fitted StandardScaler.
3. Pass the scaled vector through all 5 ANN models (one per horizon).
4. Apply $\text{clip}(\hat{y}, 0, 100)$ to constrain predictions to the valid OEE range.
5. Return the 5-horizon forecast array.

5.2 Model Performance Results

Five separate models were trained, one per horizon. The evaluation metrics on the held-out test set are:

5.3 Analysis of Results

The performance pattern is clear and physically meaningful:

Short horizons (30m–2h): The ANN achieves useful predictive skill. At 30 minutes, $\mathbf{R^2} = 0.84$ with MAPE of only 5.5%, meaning the model can forecast OEE within ±3.7 points on average. This

Horizon	R^2	MAE	RMSE	MAPE
30 minutes	0.840	3.66	4.64	5.5%
1 hour	0.790	4.30	5.42	6.4%
2 hours	0.648	5.92	7.22	8.7%
6 hours	−0.179	11.29	13.03	16.6%
8 hours	−0.494	12.83	14.43	18.6%

Table 3: Test-set performance across all five horizons.

is sufficient for operator decision support: if the model predicts a 5-point OEE drop in 30 minutes, the operator can investigate proactively.

Long horizons (6h–8h): R^2 becomes negative, indicating the model performs worse than simply predicting the mean. This is not a model deficiency but a fundamental limitation of feedforward ANNs: they have no internal memory. At 6–8 hours, OEE depends on events that have not yet occurred (future downtime, future quality degradation) that the model cannot see.

The negative R^2 for 6h and 8h horizons is an important finding: it demonstrates that *current-state-only* models are insufficient for long-horizon forecasting in manufacturing, and that temporal sequence models are required.

5.4 Advantages of the ANN Approach

- **No explicit feature engineering**: the ANN learns relevant representations from raw inputs.
- **Fast inference**: a single forward pass takes microseconds, enabling real-time deployment.
- **Separate models per horizon**: each horizon optimises its own weights, avoiding the trade-off inherent in multi-output models.
- **Interpretable via feature importance**: first-layer weight magnitudes reveal which input features dominate each horizon.

5.5 Limitations and Future Upgrades

Limitation 1: No temporal memory. The feedforward ANN treats each row independently. For long-horizon forecasting, past states contain predictive information (e.g., a quality decline trend over the last hour predicts future quality). This requires sequence models.

Limitation 2: Fixed architecture. All horizons share the same 64-32-16 layout. Shorter horizons may benefit from a wider, shallower network; longer horizons may need deeper networks with more capacity.

Future upgrades planned:

1. **LSTM (Long Short-Term Memory)**: replaces dense layers with recurrent gates that maintain a cell state, enabling the network to remember relevant past events over 100+ time steps.
2. **Transformer (Attention)**: applies self-attention mechanisms (Vaswani et al., 2017) to weigh the importance of different past time steps when making a prediction.
3. **TimesFM (Google, 2024)**: a foundation model for time-series forecasting pre-trained on 100 billion time points, which can be fine-tuned on manufacturing data with minimal examples.

5.6 Optimisation Engine

Beyond pure forecasting, the system includes an optimisation endpoint that performs grid search across RPM values:

$$\mathrm{RPM}^* = \arg \max_{\mathrm{RPM} \in \{300, 305, \ldots, 500\}} \mathrm{ANN}_{\mathrm{horizon}}(\mathrm{RPM}, \mathrm{state}) \tag{13}$$

The grid search evaluates the ANN at 41 candidate RPM values and returns the setting that yields the highest predicted OEE (or the lowest RPM that meets a target). This enables closed-loop control: the ANN recommends a speed setpoint, the VFD executes it, and the next prediction cycle evaluates the result.

5.7 Closed-Loop Conclusion

This project delivers a complete Physical AI closed-loop system for OEE forecasting and optimisation:

Factory I/O → PLC → Node-RED → ANN → Dashboard → VFD → Factory I/O

Endpoints: `POST /predict-oee` (future OEE at 5 horizons) and `POST /optimize-oee` (optimal RPM). Deployed via `uvicorn` with 4 workers and MySQL async backend.

The system transforms OEE from a backward-looking lagging indicator into a forward-looking leading indicator, empowering operators to make data-driven decisions before losses occur. Short-horizon predictions (30m–2h) are already actionable with the current feedforward ANN; future upgrades to LSTM and Transformer architectures will extend reliable forecasting to 8+ hours, enabling shift-level production planning and predictive maintenance scheduling. This represents a concrete step toward Industry 4.0 and AI-assisted digital twin manufacturing.